Semantic–Spatial Collaborative Perception Network for Remote Sensing Image Captioning

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***Abstract*— Image captioning is a fundamental vision–language task with wide-ranging applications in daily life. The existing methods often struggle to accurately interpret the semantic information in remote sensing images due to the complexity of backgrounds. Target region masks can effectively reflect the shape characteristics of targets and their potential interrelationships. Therefore, incorporating and fully integrating these features can significantly improve the quality of generated captions. However, researchers are hindered by the lack of relevant datasets that contain corresponding object masks. It is natural to ask the following: how to efficiently introduce and utilize object masks? In this article, we provide potential target masks for the publicly available remote sensing image caption (RSIC) datasets, enabling models to utilize the regional features of targets for RSIC. Meanwhile, a novel RSIC algorithm is proposed that combines regional positional features with fine-grained semantic information, abbreviated as S^2CPNet. To effectively capture the semantic information from image and position relationship from mask, respectively, the semantic and spatial feature enhancement submodules are introduced at the ends of encoder branches, respectively. Furthermore, the cross-view feature fusion module is designed to integrate regional features and semantic information efficiently. Then, a target recognition decoder is developed to enhance the ability of model to identify and describe critical targets in images. Finally, we improve the caption generation decoder by adaptively merging textual information with visual features to generate more accurate descriptions. Our model achieves satisfactory results on three RSIC datasets compared with the existing method. The related datasets and code will be open-sourced in https://github.com/CVer-Yang/SSCPNet.**

***Index Terms*— Attention mechanism, cross view, image captioning, remote sensing.**

I. Introduction

THE goal of remote sensing image captioning task is to translate the content of given remote sensing images into text [1], [2], enabling humans to intuitively grasp the crucial information contained in the images. This task combines computer vision and natural language processing, which holds significant application value in various fields [3], [4], such as intelligent security, military intelligence collection, geographic information updating, and so on.

Currently, remote sensing image caption (RSIC) faces the following challenges.

1) *Limited Visual Feature Expression Ability:* Remote sensing images are typically captured from an overhead perspective, encompassing numerous objects within a confined area. The complexity of backgrounds and the small account of targets make it challenging for models to accurately extract target features. In addition, the existence of objects with various scales weakens the ability of model to analyze the relationships between different geospatial targets. This further hinders the model from generating precise captions. Therefore, models need to accurately extract semantic features of targets and understand the spatial relationships between geospatial entities.
2) *Semantic Confusion in Description Generation:* Both the accuracy of visual content and the fluency of the generated description are key factors in evaluating the quality of the generated captions. Therefore, it is essential to efficiently utilize both visual and textual features in the text generation process. However, structural and semantic discrepancies between cross-modal data weaken the ability of model to fully leverage these features. Consequently, effectively integrating visual and text features to produce satisfactory language descriptions remains a critical challenge.

Inspired by Qu et al. [5] that introduced the RSIC task, many studies are then proposed. Considering the challenge that models cannot accurately extract target features from complex remote sensing images, researchers employ some strategies to alleviate this problem, such as multiscale feature integration [6], [7] and the introduction of attention mechanisms [8], [9]. As shown in the top of Fig. 1, these methods primarily focus on extracting fine-grained features from images and feeding them into decoders to produce captions. Different from the above methods, we improve on the following three aspects to generate high-quality captions.

First, since fine-grained features cannot fully capture the overall characteristics of targets, which makes it difficult for the model to accurately capture shape features and spatial relationships. In contrast to fine-grained features, regional features can offer an alternative perspective by focusing on the overall characteristics of significant targets. Consequently, in natural image captioning, numerous researchers combine

Received 29 August 2024; revised 1 November 2024; accepted 14 November 2024. Date of publication 20 November 2024; date of current version 3 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U21B2041, Grant 62471394, and Grant 62301385. *(Corresponding author: Qiang Li.)*

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Digital Object Identifier 10.1109/TGRS.2024.3502805

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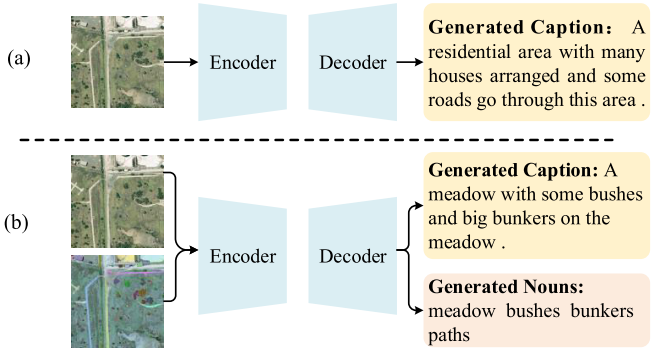


Fig. 1. Motivation of the proposed S^{2}CPNet. (a) Existing RSIC algorithms primarily rely on an encoder–decoder framework, where the encoder extracts visual features from the image and then feeds them into a language decoder to generate corresponding textual descriptions. (b) Unlike these methods, we propose a novel algorithm that takes the image and its corresponding target mask as inputs. The extracted comprehensive visual features are fed into a dual-branch decoder, which separately generates the categories of the ground targets and the textual descriptions.

fine-grained features with regional features to produce descriptions. However, the application of regional features in RSIC faces significant challenges due to the lack of corresponding annotated datasets. Creating these datasets is labor-intensive and costly. Although object detection annotations can provide the location of targets, they also may introduce extraneous noise, such as background elements, which hinder the generation of precise semantic descriptions. Compared with object detection annotations, object masks can offer a more accurate representation of target locations and shapes. Therefore, the efficient acquisition of target segmentation annotations is crucial for generating visual features with stronger expression ability.

Second, designing auxiliary task [10] is an effective approach for models in learning more robust feature representations. In the fields of RSIC, the existing multitask models mainly enhance feature extraction capabilities by incorporating scene classification task. Indeed, scene classification task can facilitate the learning of global image features. However, this task cannot guide the model to capture local details of images. On the other hand, RSIC tasks require models to capture the relationship between the geospatial targets and their corresponding words, and the scene classification task cannot provide this information. Therefore, we introduce the multitarget classification task, as shown in the bottom of Fig. 1. By introducing this task, the model can achieve an efficient alignment of targets with textual features.

Third, during the caption generation stage, most methods utilize Transformer for text modeling and employ cross Transformer to facilitate interactions between visual and textual features. However, this structure also introduces confusion in the description generation process. Specifically, in generated descriptions, some crucial entity nouns are directly related to the image content, such as “airplane” and “building.” In contrast, conjunctions, such as “and,” can be inferred simply without introducing visual features. In this case, using cross-attention mechanism to model meaningless words with visual features is unhelpful and may even impair the ability to align cross-modal features. Therefore, it is essential to design an adaptive fusion mechanism that efficiently utilizes features from different models and filters out irrelevant information, which can contribute to generating precise and coherent descriptions.

To alleviate these aforementioned challenges, we extend the existing RSIC dataset and design a novel RSIC algorithm. In terms of dataset, we combine the segment anything model (SAM) [11] to automatically generate target masks corresponding to potential targets, which can provide the overall contour and location information of the crucial targets. We also extract the entity nouns in the caption to form a key noun sequence. In terms of algorithm, we propose an RSIC method that combines the target mask and fine-grained semantic information. First, the remote sensing image and the corresponding coarse-grained annotation are taken as the input of the model, and the key feature representation of remote sensing image targets is enhanced by designing a semantic and spatial enhancement submodule. Then, a cross-view feature fusion (CVFF) module is designed to effectively fuse the fine-grained semantic information of the image with the corresponding regional features. In addition, we introduce a multitarget classification decoder to enhance the understanding ability of remote sensing images. Finally, we propose a gated-guided Transformer decoder, and it can generate linguistically fluent and correct descriptions based on textual and visual features adaptively. The main contributions of this work are summarized as follows.

1) We provide pixel-level annotations of remote sensing images for the potential targets based on publicly available RSIC datasets. These datasets form the foundation for the model to generate accurate descriptions by combining region features with fine-grained semantic features. In addition, we extract the nouns in the caption to form a sequence of categories.
2) The spatial and semantic enhancement submodules are designed at end of encoder, which can realize the feature enhancement of potential target semantic information and location features. Meanwhile, a CVFF module is designed to realize the full interaction between semantic information and regional features. It provides rich and complete visual features for the decoder.
3) The multitarget classification auxiliary task is introduced to enhance the ability of model to understand potential targets, which can align the visual features with the crucial nouns. Meanwhile, we propose a gated-guided caption decoder, which can adaptively utilize the visual features with the generated descriptions to help the model produce accurate and fluent text descriptions.

The rest of this article is organized as follows. Section II provides related work in natural image captioning and RSIC. In Section III, we describe the details of processed dataset and our proposed S^{2}CPNet. The experimental results are analyzed in Section IV. Finally, Section V offers the conclusion.

II. Related Work

In this section, we review the existing natural image captioning and the RSIC methods.

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A. Natural Image Captioning

Image captioning task aims to generate textual description that corresponds to the content of images. With the advancement of deep learning technique [12], [13], many researchers combine neural network techniques to generate captions. Vinyals et al. [14] use convolution network to extract image features and employ LSTM network to generate captions. On this basis, Xu et al. [15] integrate an attention mechanism during decoding, enabling the model to dynamically focus on relevant visual aspects. Yao et al. [16] further enhance visual feature extraction by using graph neural networks to model relationships between objects. Integrating textual information related to the image has proven beneficial to generate accurate descriptions. Inspired by this, Mun et al. [17] adopt text retrieval methods to guide the model in identifying key visual elements. Similarly, Lu et al. [18] introduce an adaptive attention network to effectively concentrate on target-related features during caption generation.

On the other hand, Anderson et al. [19] propose a bottom–up and top–down visual attention mechanism, which enables the model to locate attention targets and object regions accurately. Huang et al. [20] extend this idea by integrating attention mechanisms into Transformer networks to filter out irrelevant visual features. Fei [21] introduce a self-supervised method to improve the ability to focus on relevant image regions. Luo et al. [22] propose a diffusion model-based approach for caption generation. The method leverages image retrieval to obtain semantic information, and this information is then fed into the diffusion model to generate corresponding sentences.

Current large models show great advantages in vision–language tasks. Li et al. [23] develop a multimodal encoder–decoder structure that effectively aligns pretrained visual and textual feature encoders. They [24] introduce the query Transformer (Q-Former) to optimize the alignment of visual and verbal features, achieving strong performance across various visual–verbal tasks. Recognizing that conventional image description algorithms often require extensive data and parameters for optimal learning, Ge et al. [25] propose a retrieval-based approach that generates high-quality descriptions without extensive training, demonstrating robust domain adaptation capabilities. Although the above methods can obtain satisfactory results in natural scenes, they also exhibit certain limitations in RSIC tasks. Compared with natural images, remote sensing images contain more targets, requiring the model to have a stronger understanding of the feature targets. In addition, the model needs to accurately capture the positional relationships between the targets to generate comprehensive captions.

B. Remote Sensing Image Captioning

Inspired by NIC methods, numerous RSIC methods have emerged [26]. Unlike natural images, remote sensing images contain lots of objects with varying orientations. Therefore, models not only need to effectively identify objects within images, but also accurately capture the relationships between them. Considering the difficulty in extracting visual features, Huang et al. [6] propose a multiscale feature aggregation network and design denoising operations to eliminate redundant information from the aggregated features. Yuan et al. [8] present an innovative approach that combines a multilevel attention mechanism with graph neural networks to model image features. Similar methods have [7] and [9]. Zhang et al. [27] extract both global and local features and combine these features to generate accurate caption. To address the challenges that Transformer network does not well in capture local information, Meng et al. [28] design an attention gating mechanism to achieve precise aggregation of semantic and regional features, and a group attention mechanism is incorporated in the decoder to enhance the extraction of local features. Meanwhile, Du et al. [29] introduce a novel deformable attention mechanism to explore the relationship between foreground objects and background features.

Designing appropriate auxiliary tasks can assist models in extracting robust visual features. Zhao et al. [30] offer a network that can generate pixel-level segmentation results for key targets and image descriptions simultaneously. To facilitate multilevel interpretation of remote sensing images, Zhao et al. [31] provide a dataset that includes semantic segmentation, instance segmentation, and textual descriptions. They also design a joint optimization model for multitask cooperative training. Ye et al. [32] introduce a multilabel classification task to incorporate prior knowledge and then feed it into the decoder to enhance the accuracy of generated captions. Kandala et al. [33] introduce the image scene classification task to guide the model in generating robust visual features. In view of the lack of explainability of RSIC methods, Wang et al. [34] divide the RSIC task into word prediction and word order tasks. It performs a multiclassification task in the image to predict potential words and further generates captions by word sorting.

Combining regional features can yield high-quality visual representations. Meng et al. [35] utilize ResNet and Fast-RCNN networks to extract corresponding scene-level and object-level features from images. They then employ a graph neural network to model the correlations between different objects, enabling the generation of coherent descriptions. Furthermore, Zhao and Xiong [36] manually annotate target regions in the RSIC dataset and improve the decoder to enhance the efficiency of model in utilizing fine-grained and regional features. Large language model (LLM) technology offers significant advantages in the field of natural language processing. Hu et al. [37] propose a large remote sensing model for RSIC and remote sensing visual question answering tasks, and they also provide a fine-grained RSIC dataset. Yang et al. [38] design a two-stage vision–language training method aimed at aligning visual features with textual information. They align the features of remote sensing images and text and subsequently input the visual features into an LLM to generate corresponding textual descriptions.

Different from these approaches, we take remote sensing images with corresponding segmentation masks as inputs and, thus, can obtain richer visual features. Meanwhile, an multitarget classification task and improved decoder are introduced to fully utilize the visual and image text. The proposed method can obtain high-quality captions.

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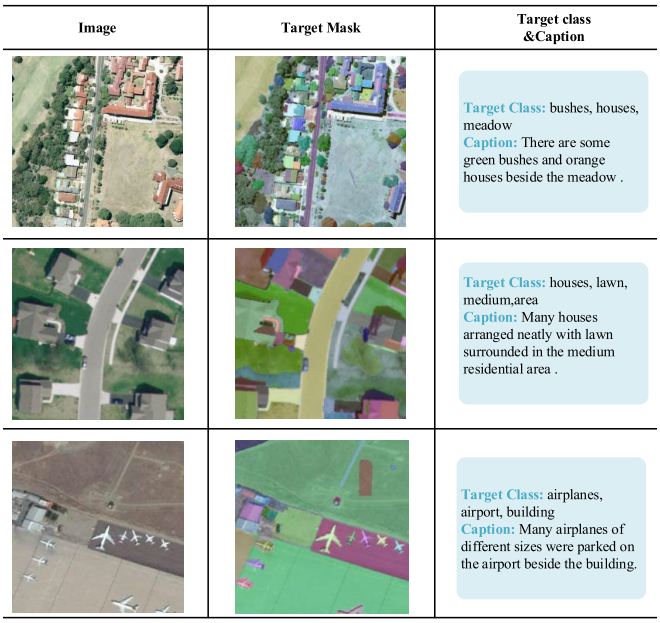


Fig. 2. Expanded Sydney caption, UCM caption, and NWPU caption datasets. These datasets comprise the original remote sensing images, potential target masks, captions for each image, and corresponding sequences of target nouns.

III. Proposed Dataset and Method

In this section, we describe the processing of the dataset and then introduce the overall structure of the designed semantic–spatial collaborative perception RSIC model with the details of each module.

A. Dataset Presentation

As shown in Fig. 2, we further expand the caption dataset to provide segmentation mask of potential targets and generate target class sequence by extracting meaningful nouns from the caption. Then, we describe the details of the dataset process.

1) Potential Target Mask Generation: To capture the positional relationship and overall shape features of potential targets in images, we provide a corresponding segmentation mask for each remote sensing image in an automated form. Specifically, the remote sensing image is input into the ViT_h model of the SAM, which produces the corresponding mask of potential objects. The generated segmentation mask is then combined with the input image. By combining images with masks, the generated image effectively represents the shape information of the target and the relative relationships between objects.

2) Multitarget Class Sequence Generation: Considering that the target masks lack relevant semantic information, we provide each remote sensing image with a corresponding target category sequence to help the model understand the semantic information of the image. Specifically, we use NLTK [39] techniques to extract nouns from the captions. However, due to semantic ambiguity, some of the extracted nouns may be meaningless. To address this problem, we implement a manual filtering process to eliminate nouns with a word frequency lower than five, as well as any meaningless nouns, such as "one" and "size." By introducing a multitarget classification task, the model can effectively align visual features with key word features, which guarantees the accuracy of the generated caption.

B. Overview of S^2CPNet

The overall structure of the proposed model is illustrated in Fig. 3. The model comprises six key components: dual-view encoder, semantic feature enhancement submodule, spatial feature enhancement submodule, CVFF module, target categorization generation decoder, and gated-guided caption generation decoder. The remote sensing images and their corresponding coarse segmentation mask are processed by a weight-sharing ResNet50 [40] network, which serves as the encoder for image feature extraction. The fine-grained semantic features and potential target spatial features are then input to the semantic and spatial feature enhancement module to strengthen the expression ability of semantic information and target location features. Subsequently, the enhanced visual features are input to the CVFF module, which facilitates the full fusion of fine-grained semantic features and location information. Finally, the aggregated features are input into the target category generation decoder and the gated-guided caption generation decoder. These decoders generate the corresponding classification categories and comprehensive descriptions of the input image.

C. Semantic–Spatial Feature Enhance

To capture complete visual features, the remote sensing image with the corresponding segmentation mask is utilized as input and fed into ResNet50 network. In each branch, the model can generate five feature maps with different resolutions. Feature map E_5 extracted by the remote sensing image feature extraction branch contains accurate semantic information, while the mask feature extraction branch captures rich object contour features M_5. To strengthen the semantic and position information conveyed by these features, we introduce a semantic enhancement submodule after the remote sensing image branch, which aims to reinforce the semantic expression. Simultaneously, a spatial enhancement submodule is implemented after the mask branch to enhance the contour and location information of the potential target.

The structure of the semantic feature enhancement submodule is illustrated in Fig. 3. Initially, a global AvgPooling operation is adopted to compress the position information of the feature E_5 and generate global visual information. Then, a sequence of operations comprising convolution, ReLU, and another convolution is applied to the generated features to eliminate redundant information. The Sigmoid function is then utilized to obtain the weights for each channel. The semantic enhancement of the feature map F_{sem} is achieved by multiplying the input semantic features by these channel weights. The overall process is defined as follows:

$$W_{\text{sem}} = \delta(\text{Conv}(\text{ReLU}(\text{Conv}(\text{AvgPool}(E_5))))) \quad (1)$$

$$F_{\text{sem}} = W_{\text{sem}} \otimes E_5 \quad (2)$$

where AvgPool$(\cdot)$ represents AvgPooling operation, Conv$(\cdot)$ represents convolution operation, $\delta(\cdot)$ represents Sigmoid operation, and $\otimes$ represents elementwise multiplication.

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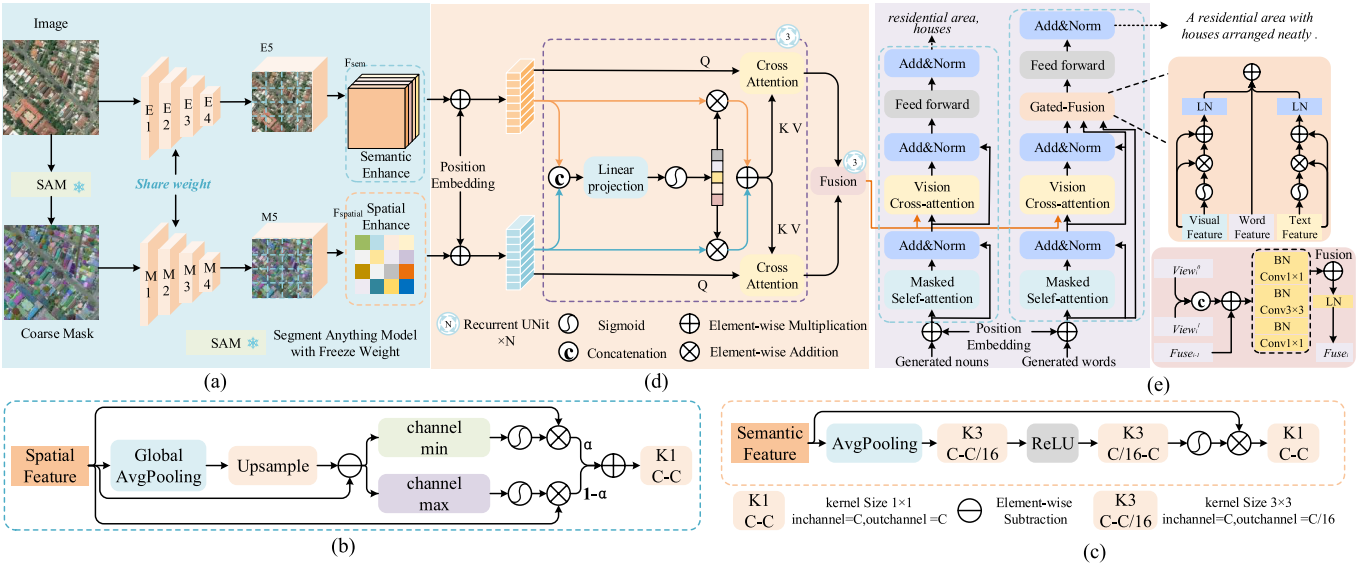


Fig. 3. Overall framework of the proposed S^{2}CPNet, and it is divided into some key parts. (a) Encoder, (b) semantic enhance submodule, (c) spatial enhance submodule, (d) cross-view feature fusion, (e) category generation decoder, and (f) gated-guided caption generation decoder.

The structure of spatial feature enhancement submodule is illustrated in Fig. 3. Initially, the low-frequency feature vectors of the feature map are obtained through an AvgPooling operation, followed by an upsampling operation to restore the resolution of M_5. Next, the high-frequency features of the feature map are acquired through a subtraction operation. Furthermore, the channel information of the high-frequency feature map is compressed using both channel min and channel max operations. Sigmoid operations are applied on these feature maps to obtain the corresponding position weights. These positional weights are then multiplied with the input feature maps to achieve positional weight. To fuse these features, a trainable parameter α with an addition operation is introduced, and the fused features are finally passed through a convolution operation to further aggregate the positional information and obtain F_{spatial}. The overall process is defined as follows:

$$W_{\text{back}} = \delta(\min(M_5 - (\text{Up}(\text{pool}(M_5))))) \tag{3}$$

$$W_{\text{fore}} = \delta(\max(M_5 - (\text{Up}(\text{pool}(M_5))))) \tag{4}$$

$$F_{\text{spatial}} = \text{Conv}(\alpha * W_{\text{back}} \otimes M_5 + (1-\alpha) * W_{\text{fore}} \otimes M_5) \tag{5}$$

where pool$(\cdot)$ represents global AvgPooling, min$(\cdot)$ represents channel min operation, and max$(\cdot)$ represents channel max operation.

AvgPooling and reshape operations are performed on the refined feature map to generate features $V_1 \in \mathbb{R}^{196\times 2048}$ and $V_2 \in \mathbb{R}^{196\times 512}$, respectively. Subsequently, the learnable 2-D position embedding algorithm is employed to incorporate positional information into these feature maps.

D. CVFF Module

The feature maps generated by the encoder provide different information. To integrate these features effectively, we propose a CVFF module that employs a coarse-to-fine feature fusion strategy. By incorporating this module, semantic information is effectively fused with positional features to generate a comprehensive visual feature representation.

The structure of the CVFF module is illustrated in Fig. 3, and it consists of three recurrent units of the same structure. Initially, the feature maps V_1 and V_2 are concatenated along the channel dimension to obtain the coarse aggregated feature. A linear operation is then applied to compress the channel dimension, facilitating crucial interactions between the input features. Subsequently, a Sigmoid function combined with channel splitting is employed to obtain positional weights for each input feature map. These input feature maps are then multiplied by their respective positional weights to enhance the positional information. Finally, the weighted feature maps are summed to generate fused feature maps with intermediate granularity. To further enhance the integration of visual information across different feature maps, we employ a cross-attention mechanism to facilitate comprehensive interaction between input feature and the fused feature. Specifically, the visual features V_1 and V_2 serve as query, and the aggregated feature map acts as key and value. This approach enables the model to capture complementary visual features from the fused maps and minimize the influence of irrelevant features. Moreover, skip connection is implemented to accelerate the convergence of model. In this way, we obtain three sets of feature maps with precise semantic information and complete structural features, respectively.

Finally, a concatenation with ResBlock is employed to merge these features. The ResBlock is composed of convolution with a kernel size of 3 × 3 and batch normalization operations. Specifically, the features from the different views View_i^0 and View_i^1 are initially aggregated using a concatenation operation and fused with the aggregated feature map Fuse_{i-1} of the previous stage using an addition operation. Then, the fused feature is input to a ResBlock block to generate the aggregated features Fuse_i of the current stage. Through these

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operations, the model can effectively fuse semantic information with spatial details and obtain complete visual feature representations.

E. Target Category Generation Decoder

Accurately recognizing the target category in an image is crucial for generating high-quality captions. The SAM model can produce the region and specific contours of the target effectively. However, the generated mask lacks corresponding semantic information, which poses challenges in accurately predicting the category of the target. To address this issue, we introduce a multitarget classification task. This task guides the model to correctly identify the target category in remote sensing images through a classification loss, thereby enhancing the comprehension of visual features. The structure of the module is illustrated in Fig. 3. First, we employ the Word Embedding algorithm [41] to encode the sequence of noun features. Subsequently, a self-attention mechanism is applied to encode noun sequences, which enables the model to understand the relationships between targets. Then, the noun features and fused visual features generated by the CVFF are fed into a cross-attention mechanism. Here, the noun sequence features serve as query, and the fused visual feature maps act as the key and value. This approach allows the model to correlate the image content with relevant noun sequences and predict the categories of targets present in the image.

F. Gated-Guided Caption Generation Decoder

Some words in the caption are inherently related to other words in the caption and require minimal visual content for accurate reasoning, such as conjunctions. In contrast, some words have a significant correlation with the visual content of an image. However, these words are fed into the cross-attention mechanism to interact with the visual features, which cause part of the modeled information being useless or even harmful to the generation of high-quality captions. Therefore, we design a gated-guided caption generation decoder. The structure of the caption decoder is shown in Fig. 3. Similar to the target category generation decoder, the input word F^i_{Word} is embedded by using the Word Embedding algorithm and a self-attention mechanism. Subsequently, the fused feature maps and textual feature are fed into the cross-attention mechanism. This decoder performs a Sigmoid operation on the text features F^i_{Text} and visual features F^i_{Visual} to determine the importance of cross-modal features for the current inference word. The feature maps are then multiplied accordingly. Finally, an addition operation followed by a linear operation is used to fuse these features. The overall process is defined as follows:

$$F^i_{\text{Caption}} = \text{LN}\left(\delta\left(F^i_{\text{Visual}}\right) \otimes F^i_{\text{Visual}}\right) + \text{LN}\left(\delta\left(F^i_{\text{Text}}\right) \otimes F^i_{\text{Text}}\right) + F^i_{\text{Word}} \tag{6}$$

where LN is the LayerNorm operation and i represents the ith word in the caption. This module allows the model to adaptively utilize both image and text features to improve the ability to generate accurate and fluent captions.

G. Loss Function

The RSIC can be reviewed as a temporal word classification task. We adopt the binary cross entropy as the loss function. The definition of this loss function is shown below

$$\mathcal{L}_{\text{cap}} = -\sum_{t=1}^{T} \log\left(p_\theta\left(s^*_t | s^*_{1:t-1}\right)\right) \tag{7}$$

where $s^*_1, s^*_2, \ldots, s^*_{t-1}$ represent the label words in caption. Similarly, the binary cross entropy is adopted as the loss function L_{word} for the multitarget classification task. The total loss function of the proposed model is defined as follows:

$$L_{\text{loss}} = L_{\text{caption}} + L_{\text{word}}. \tag{8}$$

IV. Experiments

In this section, we first introduce the datasets and evaluation metrics used by the algorithm, followed by the specific details of the algorithm. At the same time, we analyze the performance of the algorithm on three datasets and provide some visualization examples. Finally, the results of the ablation experiments are analyzed.

A. Datasets

To verify the effectiveness of this algorithm, we conduct experiments on Sydney, UCM, and NWPU datasets. These datasets come from remote sensing image scene classification tasks. The Sydney dataset contains 613 remote sensing images across seven scene categories. The UCM caption dataset includes 21 scene categories, with each category represented by 100 images at a resolution of 256 × 256 pixels. NWPU is the largest dataset for RSIC that covers 45 scene categories and includes a total of 31 500 high-resolution remote sensing images, each with a resolution of 256 × 256 pixels. All the above datasets are divided into training set, validation set, and test set according to the original division.

B. Evaluation Metrics

In this article, we choose bilingual evaluation understudy (BLEU1)-4, metric for evaluation of translation with explicit ordering (METEOR), recall-oriented understudy for gisting evaluation (ROUGE_L), consensus-based image description evaluation (CIDEr), and Sm metrics to evaluate the quality of generated descriptions.

Bilingual Evaluation Understudy: The BLEU metric is usually used in the field of machine translation. It calculates the score by comparing the overlapping n grams between the generated text and the references.

Metric for Evaluation of Translation With Explicit Ordering: It combines exact word matching with near-synonym matching and takes into account the information of word order for a more comprehensive evaluation of the similarity between the generated text and the reference text.

Recall-Oriented Understudy for Gisting Evaluation: ROUGE_L is commonly used for text summarization that focuses on the content overlap between the generated text and the reference text.

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TABLE I

EXPERIMENTS RESULTS ON THE SYDNEY DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND-BEST RESULTS. THE ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED

Method	BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr↑	Sm↑
VLAD-LSTM [42]	0.4913	0.3412	0.2760	0.2314	0.1930	0.4201	0.9164	0.4402
Sound-a-a [43]	0.7093	0.6228	0.5393	0.4602	0.3121	0.5974	1.7477	0.7794
Multimodal [5]	0.6980	0.6130	0.5440	0.5050	0.3610	0.6370	2.2020	0.9262
Soft Attenttion [42]	0.7128	0.6239	0.5527	0.4924	0.3675	0.6913	2.0343	1.0471
Hard Attention [42]	0.7689	0.6613	0.5840	0.5170	0.3719	0.6842	1.9863	0.9541
TCE loss [44]	0.7937	0.7304	**0.6717**	**0.6193**	**0.4430**	0.7130	2.4042	1.0449
Word-Sentence framework [34]	0.7891	0.7094	0.6317	0.5625	0.4181	0.6922	2.0411	0.9285
GVFGA+LSGA [27]	0.7681	0.6846	0.6145	0.5504	0.3866	0.7030	2.4522	1.0231
SVM-D BOW [45]	0.7787	0.6835	0.6023	0.5305	0.3797	0.6992	2.2722	0.9704
SVM-D CONC [45]	0.7547	0.6711	0.5970	0.5308	0.3643	0.6746	2.2222	0.9480
MLCANet [46]	**0.8310**	**0.7420**	0.6590	0.5800	0.3900	0.7110	2.3240	1.0012
MLAT [47]∗	0.7768	0.7034	0.6422	0.5862	0.3882	**0.7167**	2.3605	1.0129
Post-processing [48]	0.7837	0.6985	0.6322	0.5717	0.3949	0.7106	**2.5553**	**1.0581**
RS-CapRet [49]	0.7870	0.7000	0.6280	0.5640	0.3880	0.7070	2.3920	1.0127
S^2CPNet	0.7770	0.7208	0.6597	0.5998	0.4023	0.7163	2.4787	1.0492

TABLE II

EXPERIMENTS RESULTS ON THE UCM DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND-BEST RESULTS. THE ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED

Method	BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr↑	Sm↑
VLAD-LSTM [42]	0.7016	0.6085	0.5496	0.5030	0.3464	0.6520	2.3131	0.9536
Sound-a-a [43]	0.7484	0.6837	0.6310	0.5896	0.3623	0.6579	2.7281	1.0845
Multimodal [5]	0.7100	0.5980	0.5530	0.4600	0.3430	0.6620	2.9260	1.0977
Soft Attenttion [42]	0.7454	0.6545	0.5855	0.5250	0.3886	0.7237	2.6124	1.0624
Hard Attention [42]	0.8157	0.7312	0.6702	0.6182	0.4263	0.7698	2.9947	1.2023
TCE loss [44]	0.8210	0.7622	0.7140	0.6700	0.4775	0.7567	2.8547	1.1897
Word-Sentence framework [34]	0.7931	0.7237	0.6671	0.6202	0.4395	0.7132	2.7871	1.1400
GVFGA+LSGA [27]	0.8319	0.7657	0.7103	0.6596	0.4436	0.7845	3.3270	1.3037
SVM-D BOW [45]	0.7635	0.6664	0.5869	0.5195	0.3654	0.6801	2.7142	1.0717
SVM-D CONC [45]	0.7653	0.6947	0.6417	0.5942	0.3702	0.6877	2.9228	1.1437
MLCANet [46]	0.8260	0.7700	0.7170	0.6680	0.4350	0.7720	3.2400	1.2787
MLAT∗ [47]	0.8226	0.7539	0.6959	0.6424	0.4268	0.7729	3.0826	1.2311
Post-processing [48]	0.7973	0.7298	0.6744	0.6262	0.4080	0.7406	3.0964	1.2186
Clipcap ∗ [50]	0.8213	0.7488	0.6872	0.6320	0.4232	0.7742	3.2391	1.2671
PureT [51]	0.8573	0.8020	0.7562	0.7129	0.4686	0.8201	3.4900	1.3729
RS-CapRet [49]	0.8430	0.7790	0.7220	0.6700	0.4720	0.8170	3.5480	1.3767
S^2CPNet	**0.8722**	**0.8183**	**0.7729**	**0.7302**	**0.4822**	**0.8345**	**3.5482**	**1.3987**

TABLE III

EXPERIMENTS RESULTS ON THE NWPU DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND-BEST RESULTS. THE ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED

Method	BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr↑	Sm↑
Multimodal [5]	0.7250	0.6030	0.5180	0.4550	0.3360	0.5910	1.1790	0.6402
Soft Attention [42]	0.7310	0.6090	0.5250	0.4620	0.3390	0.5990	1.1360	0.6340
Hard Attention [42]	0.7330	0.6100	0.5270	0.4640	0.3400	0.6000	1.1030	0.6267
FC-Att+LSTM [52]	0.7360	0.6150	0.5320	0.4690	0.3380	0.6000	1.2310	0.6595
SM-Att+LSTM [52]	0.7390	0.6170	0.5320	0.4680	0.3300	0.5930	1.2360	0.6567
MLCANet [46]	0.7540	0.6240	0.5410	0.4780	0.3370	0.6010	1.2640	0.6700
BUTD∗ [19]	0.8718	0.7878	0.7206	0.6669	0.4304	0.7612	1.8866	0.9362
AoANet∗ [20]	0.8755	0.7754	0.6917	0.6207	0.3907	0.7301	1.8490	0.8976
MLAT∗ [47]	0.8527	0.7674	0.7007	0.6479	0.4327	0.7496	1.8556	0.9214
Clipcap∗ [50]	0.8394	0.7421	0.6619	0.5954	0.4142	0.7385	1.7265	0.8685
PureT [51]	**0.8880**	0.8031	0.7330	0.6750	0.4232	0.7584	1.9512	0.9519
RS-CapRet [49]	0.8710	0.7870	0.7170	0.6560	0.4360	0.7760	1.9290	0.9492
S^2CPNet	0.8801	**0.8081**	**0.7470**	**0.6965**	**0.4646**	**0.7914**	**2.0392**	**0.9979**

Consensus-Based Image Description Evaluation: The metric is used to evaluate the image captioning task, which focuses on the occurrence of keywords.

Sm: This metric is obtained by calculating the average of BLEU-4, METEOR, ROUGE_L, and CIDEr, and it can assess the overall quality of the generated caption.

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C. Experimental Settings

The model is based on PyTorch 1.8 framework, and all the experiments are conducted on an NVIDIA 3090 GPU. The batch size is set to 48, and the Adam is employed as the optimizer. The initial learning rate of the model is set to 1e-4 on the Sydney and UCM datasets and 2e-4 on the NWPU dataset. The number of cross-attention units in the CVFF is set to 3. During the training process, if the METROE not increases for continuous 5 epochs, the learning rate will adjust to one-fifth the current value. The model will terminate early if the METEOR fails to increase for continuous 20 epochs.

D. Comparison With Existing Methods

1) Results on the Sydney Dataset: Table I illustrates that the proposed S^2CPNet demonstrates competitive performance on the Sydney dataset across various metrics, such as BLEU-4, ROUGE_L, CIDEr, and Sm. Notably, the TCE loss model achieves the highest BLEU-4 and METEOR scores. This success can be attributed to the small amount of the Sydney dataset and the fact that the model incorporates a truncated loss function, which helps model alleviate overfitting issues. The GCFGA + LSGA method excels in the CIDEr metric, because it integrates global and local features of remote sensing images to obtain accurate and comprehensive semantic information. By leveraging these dual perspectives, GCFGA + LSGA captures the intricate details necessary to generate high-quality captions. Compared with these methods, our S^2CPNet integrates regional and semantic features of potential targets within the images, enabling it to capture more complete and precise target features. Consequently, it achieves satisfactory performance across multiple metrics.

2) Results on the UCM Dataset: From Table II, it can be observed that our model achieves the best performance across all metrics on the UCM dataset. The PureT model also demonstrates competitive results due to its fully Transformer-based architecture for extracting visual and textual features, which effectively reduces the semantic gap between cross-modal features. Our proposed model enhances the understanding of visual features by incorporating a target category generation decoder. This decoder helps the model to better comprehend and classify visual content, facilitating the production of more accurate linguistic descriptions. In addition, our model introduces a gated-guided caption generation decoder, which optimally leverages both visual and textual features. This design allows the model to adaptively fuse different features, which can generate more accurate and coherent descriptions. By integrating these components, our model outperforms the existing methods in producing precise and contextually relevant captions.

3) Results on the NWPU Dataset: Table III indicates that our model achieves the best performance on the NWPU dataset as well. The RS-CapRet model based on LLM technology obtains the second score in metrics, such as METEOR and ROUGE_L, demonstrating the potential of LLM technology in RSIC tasks. Our proposed S^2CPNet outperforms the PureT model by achieving a 2.15% higher score in BLEU-4, indicating that the descriptions generated by our method are more precise. The consistent performance of our model across different datasets, regardless of their scale, further attests to the robustness and versatility of our approach.

E. Visualization Results

To fully showcase the advantages of this method, we select some typical remote sensing scenes with corresponding caption, as illustrated in Fig. 4. These results show that the proposed method is able to generate more comprehensive and accurate linguistic descriptions compared with the existing methods.

On the Sydney dataset, as depicted in Fig. 4(b), the Baseline model generates statements with repetitive words. In contrast, the S^2CPNet model not only produces smoother statements, but also generates more complete descriptions containing crucial elements, such as rivers, residential areas, and houses. Meanwhile, the target category decoder can produce a category sequence for the presence of the target. The enhancement can be attributed to the gated fusion attention mechanism designed in our model, which effectively leverages both image and text features. This leads to more coherent descriptions and significantly reduces the issue of word repetition.

On the UCM dataset, as shown in Fig. 4(f), the S^2CPNet model generates more complete and accuracy descriptions compared with the Baseline model. For instance, in Fig. 4(h), the Baseline model incorrectly predicts white buildings as gray, and our S^2CPNet model can accurately describe them. This improvement is due to the CVFF module integrated into our model, which acquires more comprehensive visual features, thereby enhancing the accuracy of the generated caption. In addition, the multitarget classification task helps the model recognize the feature of "baseball," "diamond," "sand," and "weed," which further enhances the ability of model to comprehend and interpret images. This capability is particularly important in complex scenes.

On the NWPU dataset, as illustrated in Fig. 4(k), the S^2CPNet model generates captions that are more comprehensive in content compared with the Baseline model. The descriptions encompass a broader range of elements and provide a clearer depiction of the scene. Furthermore, in Fig. 4(n), the Baseline model incorrectly predicts the number of basketball courts, demonstrating a significant shortcoming. In contrast, our model leverages the spatial information by introducing the corresponding mask. Therefore, the proposed S^2CPNet is more sensitive to the number of targets and produces reliable captions.

F. Ablation Study

In this section, we design ablation in the UCM dataset to demonstrate the effectiveness of the design module.

1) Only Baseline: We utilize two variants as Baselines: the first variant (Baseline1) employs ResNet50 as the encoder and a Transformer network as the decoder, while the second variant (Baseline2) replaces the standard Transformer network with a gated-guided Transformer mechanism. As shown in Table IV, the improved Transformer decoder gains improvements in BELU4, METEOR, ROUGE_L, and CIDEr metrics compared

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Fig. 4. Examples of sentences generated by S^{2}CPNet from Sydney dataset (a)–(e), UCM dataset (f)–(g), and NWPU dataset (k)–(o). The red words and blue words indicate the mistakes and advantages of our method compared with Baseline.

with Baseline1. These enhancements are attributed to the ability of model to fully leverage both visual and textual features, thereby significantly improving the accuracy of the generated descriptions. Consequently, we adopt the ResNet50 with a gated-guided Transformer as our Baseline.

2) Introducing Segmentation Masks and CVFF: Based on Baseline, we take the remote sensing image and the corresponding target segmentation mask as the input to the model and use the CVFF module to realize the fusion of these features. As shown in Table V, incorporating the

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TABLE IV
ABLATION RESULTS ON BASELINE OF MODEL ON THE UCM DATASET. THE BOLD INDICATE THE BEST RESULTS

Components				Metric							
Method	ResNet50	Transformer	Dual-Gated Transformer	BLEU-1 ↑	BLEU-2 ↑	BLEU-3 ↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr ↑	Sm ↑
Baseline1	✓	✓		**0.8589**	**0.8029**	**0.7481**	0.6942	0.4501	0.8026	3.1708	1.2794
Baseline2	✓		✓	0.8536	0.7989	**0.7481**	**0.7016**	**0.4560**	**0.8040**	**3.3007**	**1.3155**

TABLE V
ABLATION EXPERIMENTS ON EACH COMPONENTS OF MODEL ON THE UCM DATASET. THE BOLD INDICATES THE BEST RESULTS

Components						Metric							
Method	Baseline	Objcet masks	SSFE	CVFF	Nouns Predict	BLEU-1 ↑	BLEU-2 ↑	BLEU-3 ↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr ↑	Sm ↑
Model1	✓					0.8536	0.7989	0.7481	0.7016	0.4506	0.8040	3.3307	1.3217
Model2	✓	✓		✓		0.8606	0.8056	0.7606	0.7208	0.4629	0.8125	3.4428	1.3597
Model3	✓	✓		✓	✓	0.8499	0.8061	0.7688	**0.7332**	0.4662	0.8018	3.4063	1.3518
S^2CPNet	✓	✓	✓	✓	✓	**0.8722**	**0.8183**	**0.7729**	0.7302	**0.4822**	**0.8345**	**3.5482**	**1.3987**

TABLE VI
ABLATION RESULTS ON DIFFERENT FEATURE FUSION STRATEGIES ON THE UCM DATASET. THE BOLD INDICATES THE BEST RESULTS

Components						Metric							
Method	Baseline	Object masks	add	contact	CVFF	BLEU-1 ↑	BLEU-2 ↑	BLEU-3 ↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr ↑	Sm ↑
Model2-A	✓	✓	✓			0.8458	0.7835	0.7282	0.6776	0.4417	0.7971	3.2148	1.2828
Model2-B	✓	✓		✓		0.8412	0.7841	0.7315	0.6871	**0.4662**	0.8018	3.4063	1.3403
Model2	✓	✓			✓	**0.8606**	**0.8056**	**0.7606**	**0.7208**	0.4629	**0.8125**	**3.4428**	**1.3597**

TABLE VII
ABLATION RESULTS ON EACH COMPONENTS OF SSFE ON THE UCM DATASET. THE BOLD INDICATES THE BEST RESULTS

Components				Metric							
Method	Model3	Semantic Enhance	Spatial Enhance	BLEU-1 ↑	BLEU-2 ↑	BLEU-3 ↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr ↑	Sm ↑
Model4-A	✓	✓		0.8621	0.8132	0.7663	0.7214	0.4613	0.8138	3.3598	1.3339
Model4-B	✓		✓	0.8714	**0.8325**	**0.7941**	**0.7579**	0.4752	0.8264	3.4396	1.3747
S^2CPNet	✓	✓	✓	**0.8722**	0.8183	0.7729	0.7302	**0.4822**	**0.8345**	**3.5482**	**1.3987**

segmentation masks and utilizing the CVFF module to aggregate fine-grained semantic features with contour features enable the model to obtain more comprehensive visual features. Consequently, the captions generated by the model show significant improvements, with the increases of 1.92% in the BLEU4 metric and 11.21% in the CIDEr metric, respectively.

In addition, we also design different feature fusion methods to replace the CVFF module, as shown in Table VI, including feature addition (Model2-A) and feature contraction (Model2-B). However, the performance of the models all shows a decrease compared with Model2, which proves that simple fusion operations are insufficient for effectively integrating semantic and contour features.

3) Introduction of Multitarget Classification Task: As shown in Table V, based on Model2, we introduce the task of multitarget classification to achieve feature alignment between visual features and nouns. This enhancement improves the semantic comprehension of remote sensing images. As a result, the model shows an improvement of 1.24% in the BLEU-4 metric and 0.33% in the METEOR metric, indicating that the produced predictions are more consistent with the ground truth.

4) Introduction of Semantic and Spatial Feature Enhance Submodule: As shown in Table VII, based on Model3, we incorporate a semantic feature enhancement submodule and a spatial feature enhancement submodule separately. The experimental results demonstrate a 1.2% improvement in the ROUGE_L score with the inclusion of the spatial feature enhancement submodule. Furthermore, the integration of this submodule leads to consistent improvements across various evaluation metrics, with the Sm indicator showing a notable increase of 2.29%. This indicates that the overall quality of the captions generated by the model improves significantly. It is due to the fact that the spatial feature enhancement submodule can strengthen the boundary regions of objects and provide the model with sufficient target position relations, which is crucial for the RSIC task. Moreover, the combination of Model3 with both spatial and semantic feature enhancement submodules further enhances the performance of model, which demonstrates the effectiveness of this combination.

V. CONCLUSION

This article proposes an RSIC algorithm that combines fine-grained semantic information with region regions, named

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S^2CPNet. In terms of dataset, we provide masks for potential targets in the RSIC dataset, allowing the model to leverage regional features effectively. To fully align visual features with textual features, we use NLTK to extract nouns from captions and form a sequence of target categories. In terms of algorithms, the semantic enhancement and spatial enhancement submodules are introduced to enhance target semantic features and spatial features. Then, a CVFF module is developed to effectively integrate fine-grained semantic features with regional features, which can generate a comprehensive visual feature. Furthermore, the multitarget classification task is introduced to align visual features with text features. In this way, the model can accurately describe the objects in the image. Finally, we propose an improved decoder that employs gated-guided mechanism to utilize visual and textual features efficiently. The proposed approach produces accurate and fluent captions on the RSIC dataset, and ablation experiments demonstrate the necessity of these improvements.

In the future, we will focus on building a new RSIC dataset that provides both detection and segmentation of targets as well as image caption. This will further promote the development of the RSIC.

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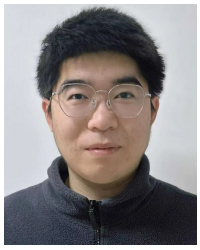
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